Udemy ML, DL, NLP
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Sec 37: Decision Tree Classifier & Regressor

Lec-7) Decision Tree Regression

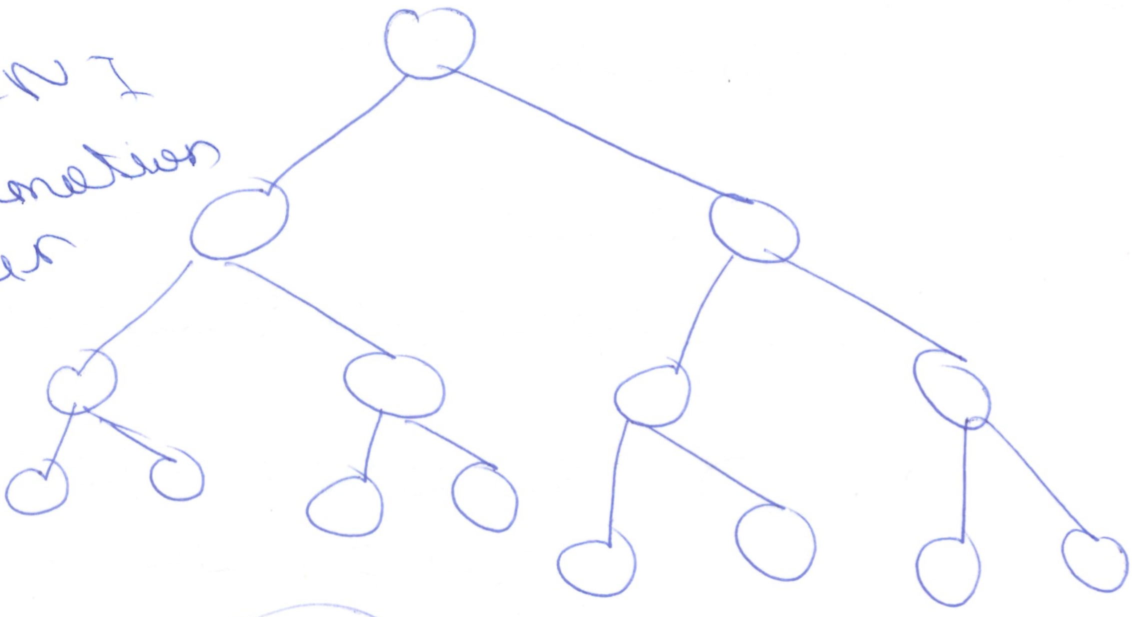
Decision Tree Regression

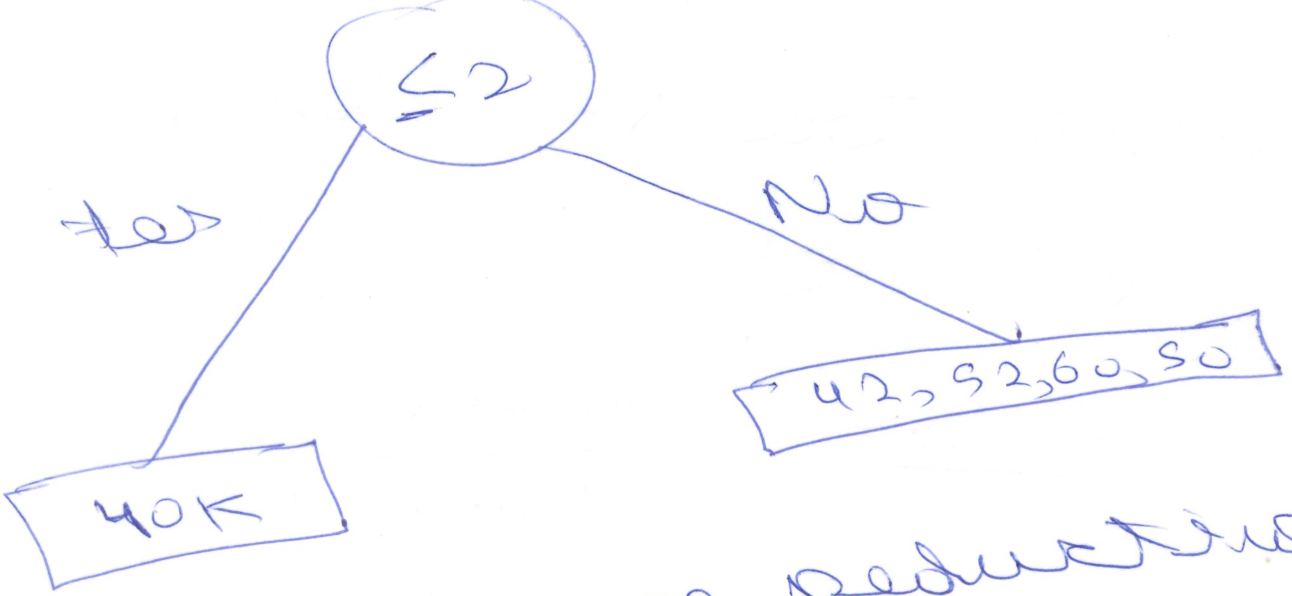
Dataset

Exp	Gap	Salary
2	Yes	40K
2.5	Yes	42K
3	No	52K
4	No	60K
4.5	Yes	56K

GINI

Information Gain





=) Variance reduction (Regr

① Variance

$$\Rightarrow \frac{1}{n} \sum_{i=1}^{n} (y - \bar{y})^2$$

Average

(3)

Variance (cost)

$= \frac{1}{5}\left[(40-50)^2 + (42-50)^2 + (42-50)^2 + (60-50)^2 + (56-50)^2\right]$

$= \frac{1}{5}\left[100 + 64 + 4 + 100 + 36\right]$

$= 60.8$

$\text{Variance}(c_1) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2$

$= \frac{1}{1} \times (40-50)^2$

$= 100$

Variance (c_2)

$= \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2$

$= \frac{1}{4}\left[(42-50)^2 + (52-50)^2 + (60-50)^2 + (56-50)^2\right]$

$$= \frac{1}{4}\left[64 + 4 + 100 + 36\right]$$ (4)

$$= 51$$

=) <u>variance reduction</u>

$$= var(root) - \sum w_i\, var(child)$$

$$= 60.8 - \left[\frac{1}{5} \times 100 + \frac{4}{5} \times 51\right]$$

$$= 60.8 - 20 - 40.8$$

$$var(c_1) = \frac{1}{2}\left[(40-50)^2 + (42-50)^2\right]$$

$$= \frac{1}{2}\left[100 + 64\right]$$

$$= \frac{164}{2} = 82$$

$$var(child) = \frac{1}{3}\left[4 + 100 + 36\right]$$

$$= \frac{140}{3} = 46.61$$

variance relation

$$= 60.8 - \left[\frac{2}{5} \times 82 + \frac{3}{5} \times 46.61\right]$$

$$= 60.8 - \left[32.8 + 27.96\right]$$

$$= +0.0041$$

Variance reduction (5)
(Left split)

< VR (Right split)

Test data

Left tree

≤ 2

≤ 2.5

40.42

52, 60, 56

5 ≤ 2.5

40, 42

52, 60, 56

Leaf Node

o/p = [40+42]
= 41